## bci2a Representation

### Intra-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.03 | +0.06 | -0.08 | -0.02 | -0.02 |
| Spectral | +0.03 | +0.03 | +0.05 | -0.07 | -0.01 | -0.04 |
| Wavelet | +0.04 | +0.03 | +0.06 | -0.08 | -0.02 | -0.03 |
| Covariance | +0.02 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.02 | +0.05 | -0.06 | -0.02 | -0.02 |
| Riemannian | +0.03 | +0.03 | +0.04 | -0.06 | -0.02 | -0.02 |
| Fused | +0.03 | +0.03 | +0.04 | -0.06 | -0.01 | -0.03 |

### Cross-session

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.03 | +0.06 | -0.08 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.04 | +0.05 | -0.06 | -0.02 | -0.04 |
| Spectral | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |

### Cross-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.03 | +0.05 | -0.07 | -0.01 | -0.02 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |

## eegmmidb Representation

### Intra-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.04 | +0.05 | -0.07 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.04 | +0.03 | +0.04 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.04 | +0.03 | +0.04 | -0.06 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.04 | +0.04 | -0.06 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |

### Cross-session

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.04 | +0.05 | -0.08 | -0.02 | -0.02 |
| Temporal | +0.03 | +0.04 | +0.05 | -0.06 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.03 | +0.04 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.07 | -0.01 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.04 | +0.05 | -0.07 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.01 | -0.02 |

### Cross-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.04 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.04 | +0.05 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.03 | +0.03 | +0.06 | -0.08 | -0.02 | -0.03 |
| Covariance | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.02 |
| Riemannian | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |

## weibo Representation

### Intra-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.04 | +0.05 | -0.06 | -0.02 | -0.03 |
| Spectral | +0.02 | +0.04 | +0.05 | -0.07 | -0.01 | -0.03 |
| Wavelet | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.02 | +0.04 | -0.06 | -0.02 | -0.02 |
| Riemannian | +0.03 | +0.03 | +0.04 | -0.06 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.04 | -0.06 | -0.01 | -0.02 |

### Cross-session

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.04 | +0.02 | +0.06 | -0.07 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Wavelet | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.02 |
| Covariance | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.03 |
| Riemannian | +0.02 | +0.04 | +0.06 | -0.07 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.06 | -0.01 | -0.03 |

### Cross-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.03 | +0.04 | +0.04 | -0.06 | -0.02 | -0.03 |
| Temporal | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.04 |
| Spectral | +0.03 | +0.04 | +0.06 | -0.07 | -0.02 | -0.04 |
| Wavelet | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |
| Riemannian | +0.04 | +0.03 | +0.05 | -0.06 | -0.02 | -0.03 |
| Fused | +0.02 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |

## zhou Representation

### Intra-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.02 | +0.02 | +0.05 | -0.06 | -0.01 | -0.02 |
| Temporal | +0.02 | +0.04 | +0.04 | -0.07 | -0.01 | -0.02 |
| Spectral | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.02 |
| Wavelet | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.04 | -0.07 | -0.01 | -0.02 |
| CSP-6 | +0.03 | +0.03 | +0.04 | -0.06 | -0.01 | -0.02 |
| Riemannian | +0.03 | +0.02 | +0.04 | -0.06 | -0.01 | -0.02 |
| Fused | +0.02 | +0.03 | +0.04 | -0.06 | -0.01 | -0.02 |

### Cross-session

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.04 | +0.05 | -0.07 | -0.01 | -0.04 |
| Wavelet | +0.04 | +0.03 | +0.06 | -0.07 | -0.03 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.03 | +0.05 | -0.06 | -0.03 | -0.03 |
| Fused | +0.02 | +0.03 | +0.06 | -0.07 | -0.01 | -0.03 |

### Cross-subject

| | 1–38 Hz | Alpha | Beta | Delta | Gamma | Theta |
|---|---|---|---|---|---|---|
| Statistical | +0.04 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Temporal | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Spectral | +0.03 | +0.04 | +0.05 | -0.07 | -0.02 | -0.04 |
| Wavelet | +0.03 | +0.03 | +0.06 | -0.07 | -0.02 | -0.03 |
| Covariance | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| CSP-6 | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Riemannian | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |
| Fused | +0.03 | +0.03 | +0.05 | -0.07 | -0.02 | -0.03 |

Δ Balanced Accuracy
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15